

Principles of Universal Design

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Dropped Kerb



- Selwyn Goldsmith, author of *Designing for the Disabled* (1963)

Skovtårnet, Haslev

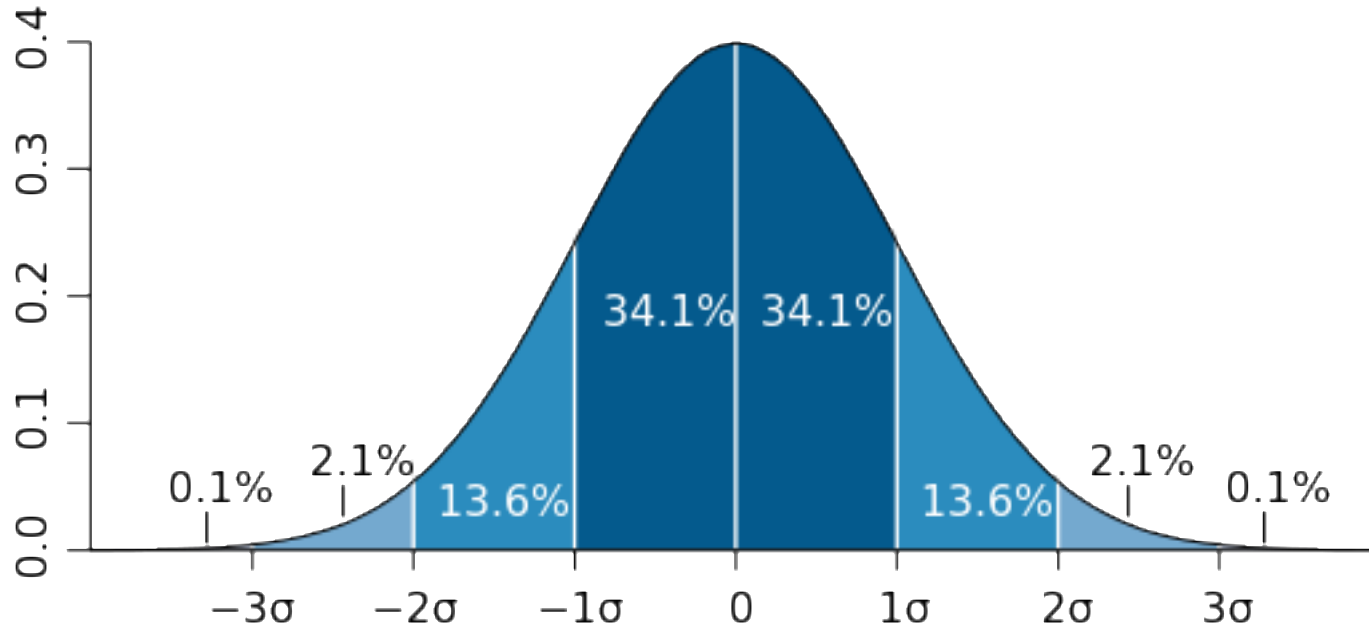


Accessibility: Tailors designs to meet the needs of individuals with disabilities, ensuring that products and environments are accessible to them.

Universal Design: Focuses on creating products and environments that are inherently accessible and usable by all people, regardless of their abilities or age.

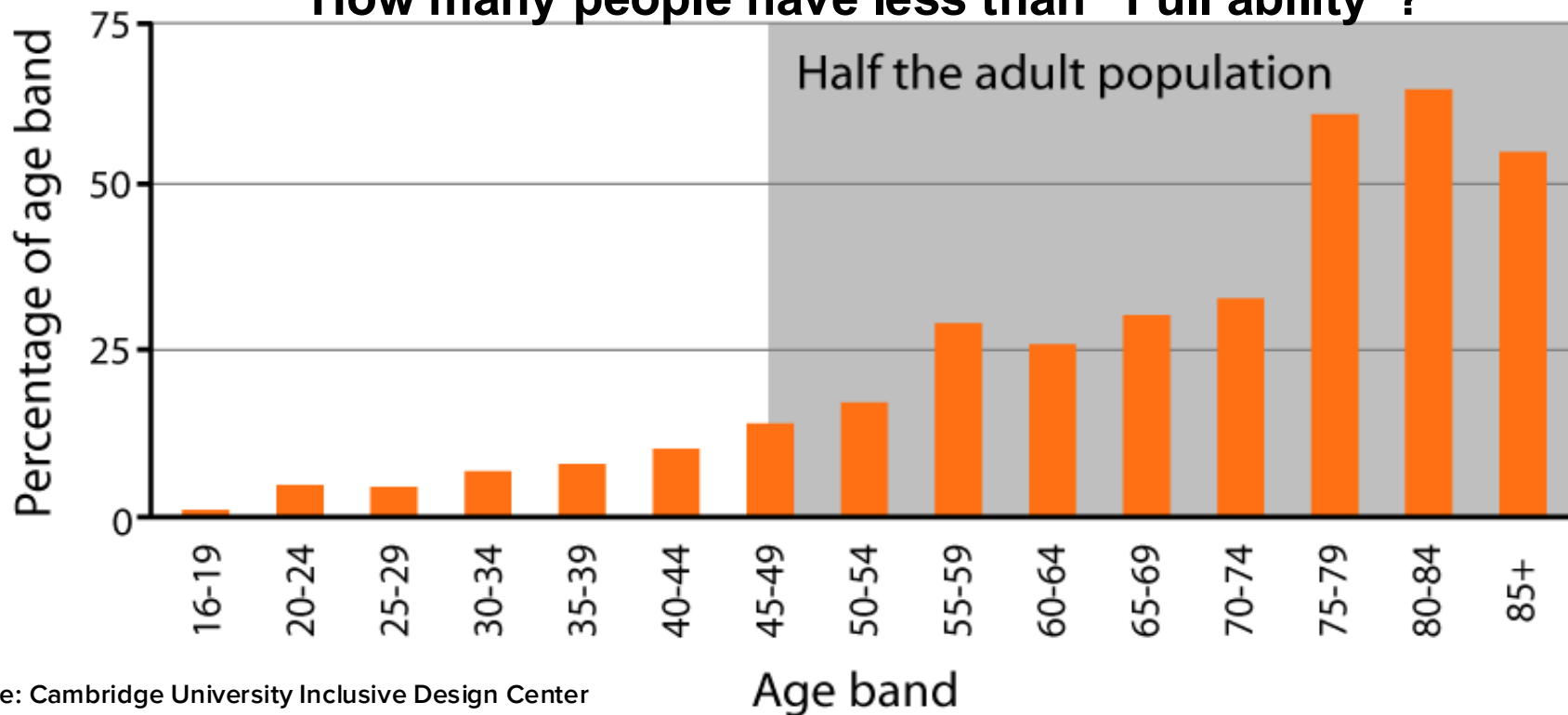


Normal distribution of body size, vision, intelligence etc.



The Impact of Age

How many people have less than “Full ability”?



Source: Cambridge University Inclusive Design Center

Exclusion

- Århus new light rail



Exclusion

- Social exclusion activates the same center in the brain as physical pain
- Provokes anxiety, uncertainty, regret, hostility, feeling of 'inappropriateness', feeling of lack of control.”
- UN's Sustainable Development Goals
 - LEAVING NO ONE BEHIND

Discrimination of senior people

Seniors often struggle with IT
.....and it is not always that the
grandchildren are able to help.



Gender discrimination

The first woman in space was many years delayed because a spacesuit for women was not developed.



The Seven Principles of Universal Design

1. Equitable Use

The design is useful and marketable to people with different abilities



2. Flexibility in use

The design accommodates a wide range of individual preferences and abilities



Right & left-handed scissors

3. Simple and intuitive use

Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level



4. Perceptible information

The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities



5. Tolerance for error

The design minimizes hazards and the adverse consequences of accidental or unintended actions



6. Low physical effort

The design can be used efficiently and comfortably and with a minimum of fatigue



7. Size and space for approach and use

Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture or mobility



Inclusive Technologies and Universal Design Driving Innovation

Electric Toothbrushes



- Created for patients with limited motor skills by Dr. Philippe Guy Woog in Switzerland in 1954
- The [global electric toothbrush market](#) was valued at \$2,200 million in 2016, and is projected to reach \$3,770 million by 2023

Hearing Aids



Ca. 1940



Ca. 1950



Ca. 1960



Now



DTU Skylab

BEVICA

DTU Management

DH
DANSKE HØRINGSORGANISATIONER

Market

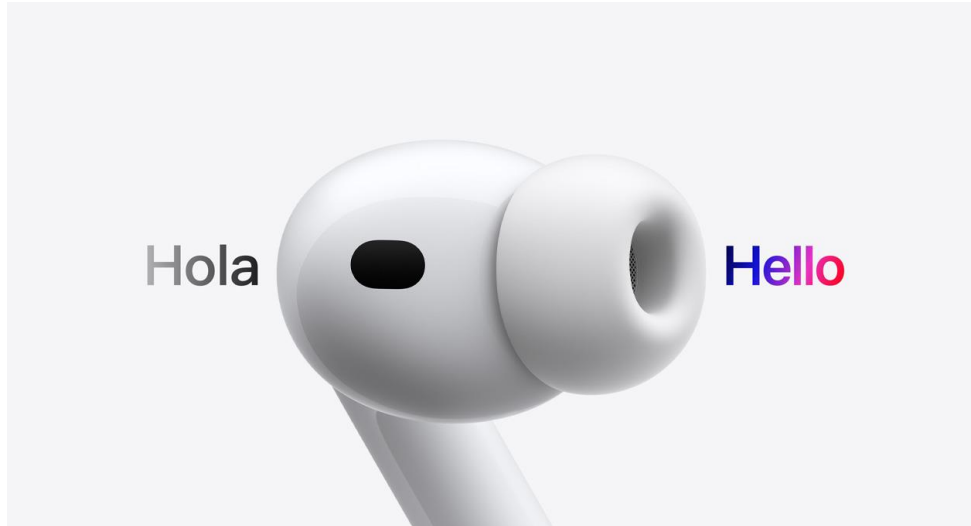
Brands

Group	Key HA Brands	Est World Mkt %	
Sonova	Phonak Unitron Hansaton	25%	DK
WDH	Oticon Bernafon Sonic	24%	
GN	ReSound Beltone Interton	16%	DK
Sivantos	Signia/Siemens Rexton	16%	
Starkey	Starkey Micro-Tech NuEar	11%	
Widex	Widex	8%	DK

Volume

- 16 million units are sold worldwide each year
- Only one in three people who need a hearing aid use one
- Untreated hearing loss costs the EU €185 billion annually, cf. <https://www.hear-it.org/untreated-hearing-loss-eu-costs-more-whole-eu-budget>

Tomorrow



Hearing Test, Hearing Aid feature,
and active Hearing Protection



Drinksaver



- DTU Design & Innovation semester project
- Plastic solution that hinders spillage of various drinks
- Developed for people with motor disabilities and wheelchair users in collaboration with Roskilde Festival
- Has found wide use among sailors, toddlers and people with motor challenges





reddot winner 2024



DTU Skylab

BEVICA

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DH

DANSKE HANDICAPORGANISATIONER

Birger Bergman Jeppesen



GazeTalk 2016



Outlook



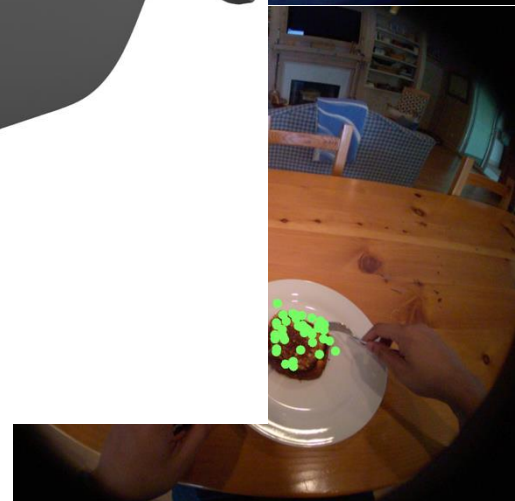
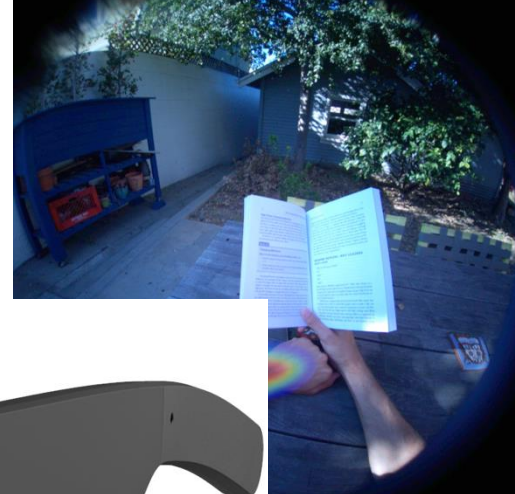
Meta Orion Glasses



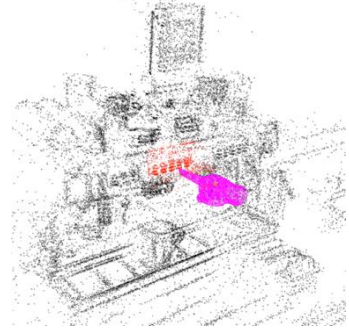
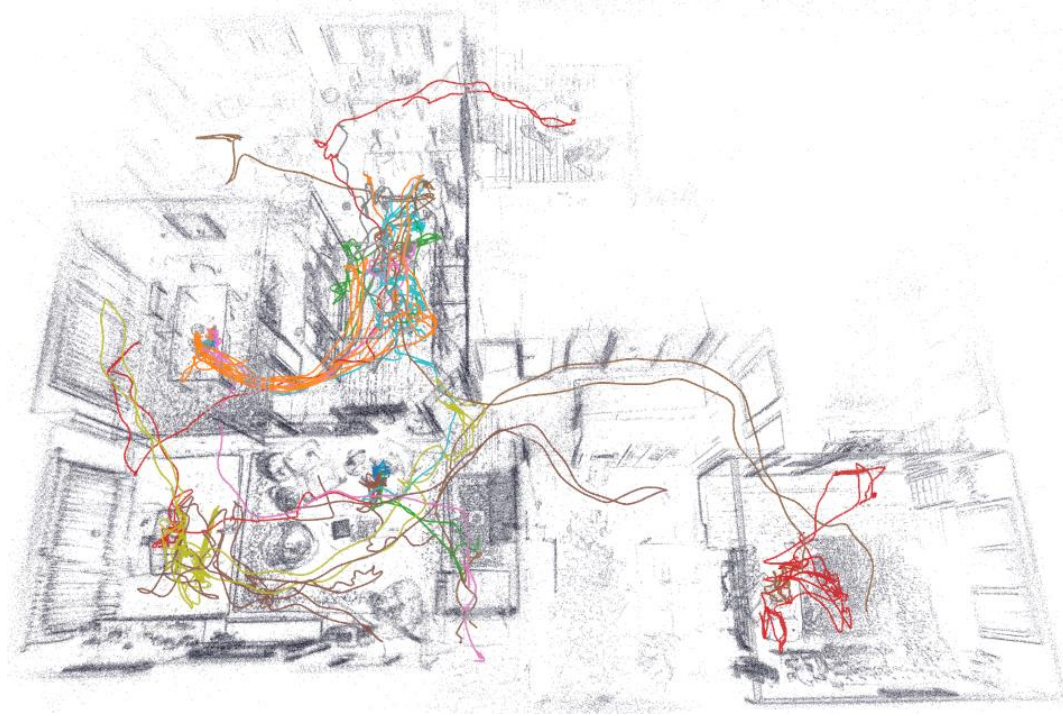
Multimodal Sensor Integration



EYE TRACKING
CAMERAS
320 X 240 PX,
LOW RES



Monitoring @home



AI to support Cognition and/or Vision challenges



Has this lectures met some of the learning goals of this course?

- Identify how social values are embedded in engineering projects and how technology can become a matter of public concern.
- Describe and evaluate the primary social processes of designing, implementing, adopting, and contesting innovation.
- Discuss key dimensions of responsible research and innovation and apply them to evaluate engineering projects, especially anticipation, reflexivity, inclusion, and responsiveness.

For instance:

- **Social values in engineering design** are illustrated through cases such as dropped kerbs, Skovtårnet, and inclusive public transport, showing how design choices can enable or exclude users.
- **Technology as a public concern** is demonstrated through examples of exclusion in light rail systems and digital barriers for seniors, highlighting the societal consequences of poor design.
- **Social processes of innovation** are shown through the evolution of hearing aids and electric toothbrushes, illustrating how assistive technologies can move from niche solutions to mainstream products.
- **Responsible Research and Innovation** is operationalised through the Seven Principles of Universal Design and applied in cases such as DrinkSaver and GazeTalk, emphasising inclusion, anticipation, and responsiveness.